

Student Performance Analytics & Decision Support System

Purpose: The Data Dictionary defines all data elements used in the analytics solution, including source, description, data type, and business meaning.

Dataset 1: students.csv

Column Name	Data Type	Description	Business Meaning
student_id	String	Unique student identifier	Identifies each student
student_name	String	Student full name	Student identification
gender	String	Student gender	Demographic analysis
class	String	Grade level	Academic grouping
section	String	Division	Class-wise analysis
age	Integer	Student age	Student profile
ability_level	String	Academic ability category	Performance segmentation
attendance_category	String	Attendance classification	Attendance analysis

Dataset 2: question_bank.csv

Column	Data Type	Description	Business Meaning
question_id	String	Unique question identifier	Question tracking
subject	String	Subject name	Subject analysis
chapter	String	Chapter name	Learning gap analysis
difficulty	String	Question difficulty	Question analysis
marks	Integer	Question marks	Score calculation

Dataset 3: tests.csv

Column	Data Type	Description
test_id	String	Unique test identifier
test_name	String	Assessment name
subject	String	Assessment subject
test_date	Date	Date of assessment
difficulty	String	Test difficulty
total_questions	Integer	Number of questions

Dataset 4: responses.csv

Column	Data Type	Description
student_id	String	Student identifier
test_id	String	Assessment identifier
question_id	String	Question identifier
score	Integer	1 correct / 0 incorrect

Dataset 5: scores.csv

Column	Data Type	Description
student_id	String	Student identifier
test_id	String	Test identifier
correct_answers	Integer	Correct responses
incorrect_answers	Integer	Wrong responses
percentage	Decimal	Test percentage
grade	String	Grade obtained
pass_fail	String	Pass status
subject	String	Subject
test_difficulty	String	Difficulty level

Dataset 6:
student_progress.csv

Column	Data Type	Description
student_id	String	Student identifier
avg_percentage	Decimal	Average score
best_score	Decimal	Highest score
worst_score	Decimal	Lowest score
first_score	Decimal	Initial performance
latest_score	Decimal	Latest performance
tests_attempted	Integer	Number of tests
improvement	Decimal	Performance change

Dataset 7:
student_risk_report.csv

Column	Data Type	Description
student_id	String	Student identifier
avg_percentage	Decimal	Average score
performance_band	String	Performance category
risk_level	String	Risk classification
intervention_required	String	Action requirement

Dataset 8: question_analytics.csv

Column Name	Data Type	Description	Business Meaning
question_id	String	Unique question identifier	Identifies individual assessment questions
subject	String	Subject name	Enables subject-wise question analysis
chapter	String	Chapter associated with question	Identifies topic-level performance
difficulty	String	Question difficulty level	Easy / Medium / Hard classification
attempts	Integer	Number of student attempts	Measures question exposure
correct	Integer	Number of correct responses	Measures successful attempts
incorrect	Integer	Number of incorrect responses	Measures learning difficulty
accuracy_rate	Decimal	Percentage of correct responses	Measures question effectiveness
performance_category	String	Question classification	Easy / Moderate / Difficult

Dataset 9: chapter_analytics.csv

Column Name	Data Type	Description	Business Meaning
subject	String	Subject name	Subject-level analysis
chapter	String	Chapter name	Topic-level analysis
attempts	Integer	Total question attempts	Measures student interaction
correct	Integer	Correct responses	Successful learning indicators
incorrect	Integer	Incorrect responses	Learning gaps
accuracy_rate	Decimal	Chapter accuracy percentage	Measures chapter understanding
performance_category	String	Chapter performance classification	Strong / Average / Weak